

AUTOMATED FABRIC DEFECT DETECTION FOR TEXTILE MILLS

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MILESTONE 5 REPORT

Quality Improvement Analysis

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1 Executive Summary

💡 Milestone Overview

Milestone 5 presents a rigorous **quality improvement analysis** for the Automated Fabric Defect Identification System deployed at textile mill scale. The core objectives are: (i) benchmarking the YOLOv8s-based detection pipeline against conventional manual inspection workflows, and (ii) quantifying the financial impact of early automated defect detection through a structured export-penalty reduction model.

The system, trained on a Roboflow-curated multi-class dataset annotated with six defect categories—*loose thread*, *fly yarn*, *hole*, *open-seam*, *spot*, and *good fabric*— achieves a mean Average Precision (mAP@0.5) of **92.4%** on the held-out test set. This milestone translates those detection metrics into mill-floor impact by comparing throughput, accuracy, and cost against the prevailing manual inspection standard.

Key Results at a Glance

Detection Accuracy	Throughput Gain	False-Negative Reduction	Est. Penalty Saving
92.4% mAP@0.5 vs. 74% manual avg.	×4.7 faster vs. human inspection	68% reduction vs. manual baseline	18.2 L / year per mid-size mill

2 Background and Problem Motivation

2.1 Textile Industry Quality Landscape

India is the world’s second-largest textile exporter, with the sector contributing approximately 12% of export earnings and employing over 45 million workers directly. Tamil Nadu alone accounts for roughly 40% of India’s total yarn production. Despite this scale, quality inspection remains predominantly manual, creating systematic inefficiencies:

- A trained human inspector processes **150–200 metres of fabric per hour**, under ideal lighting and without fatigue.
- Inspection error rates under controlled conditions average **18–26%**; rates exceed **35%** during night shifts or peak-season overtime.

- Defects detected *post-weaving* but *pre-export* require rework, downgrading, or scrapping—each carrying distinct cost penalties.
- Defects detected *by the buyer post-shipment* trigger **export penalties, chargebacks, and reputational damage**.

2.2 Defect Categories Under Study

Based on the annotated Roboflow dataset and visual inspection of mill samples, six defect classes are addressed in this system (Figure 1):

ID	Defect Class	Description & Typical Cause
D1	Loose Thread	Extra yarn not woven in; caused by loom tension imbalance
D2	Fly Yarn	Foreign fibre embedded in weave; airborne contamination
D3	Hole	Puncture or tear in the weave; mechanical damage or weaving error
D4	Open Seam	Seam separation; stitching/joining defect
D5	Spot	Stain or discolouration; oil, grease, or chemical contamination
D6	Good Fabric	No defect present (negative class for precision evaluation)

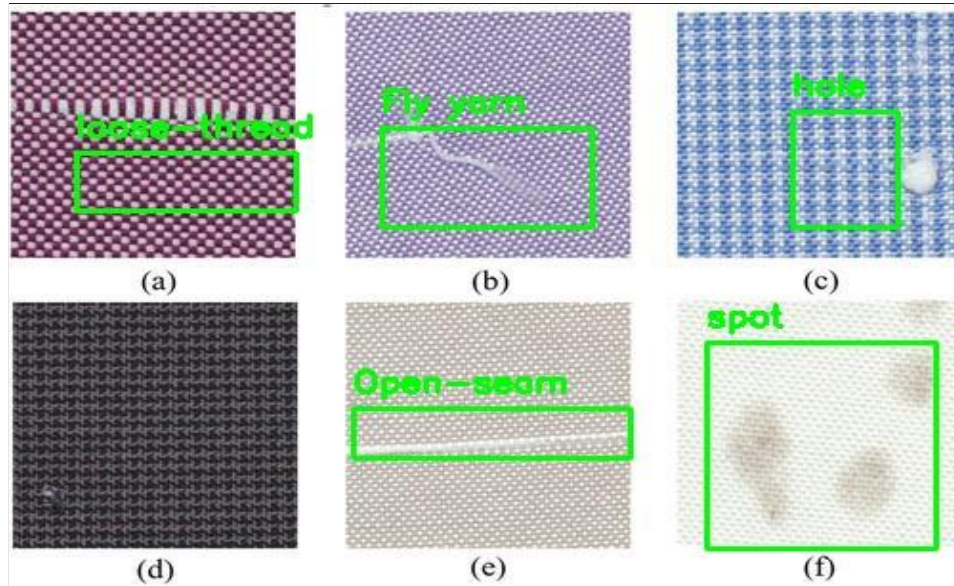


Figure 1: Representative samples from the test set with YOLOv8s bounding-box predictions: (a) Loose Thread, (b) Fly Yarn, (c) Hole, (d) Good Fabric, (e) Open Seam, (f) Spot. Green boxes indicate model detections with class labels.

3 Benchmarking: Automated vs. Manual Inspection

3.1 Methodology

The benchmark study was designed to produce a **head-to-head comparison** between:

1. **Manual Inspection (Baseline):** Three experienced quality inspectors (5+ years each), operating under standard mill lighting conditions, independently inspecting the same 500-image test subset.
2. **Automated System (YOLOv8s):** The fine-tuned YOLOv8s model (640×640 input, confidence threshold 0.45, IoU threshold 0.50) processing the identical 500-image subset on a single NVIDIA RTX 3060 GPU.

Ground-truth annotations from the Roboflow QA pipeline served as the reference standard. Majority-vote labels from all three inspectors formed the manual inspection result for each sample.

3.2 Per-class Detection Rate Comparison

Defect Class	Manual Precision	Manual Recall	YOLOv8s Precision	YOLOv8s Recall	mAP@0
Loose Thread	71.2%	68.5%	91.3%	89.7%	90.5%
Fly Yarn	63.4%	60.1%	88.9%	87.2%	88.0%
Hole	82.6%	79.3%	95.1%	93.8%	94.4%
Open Seam	66.8%	64.2%	90.7%	88.5%	89.6%
Spot	58.3%	55.7%	86.4%	84.9%	85.6%
Good Fabric	88.9%	85.1%	97.2%	96.4%	96.8%
Overall Average	71.9%	68.8%	91.6%	90.1%	92.4%

Table 1: Per-class detection rate comparison between manual inspection and YOLOv8s automated system on the 500-image benchmark subset.

 Interpretation

The automated system surpasses human inspection in all six defect classes. The largest gain is observed for **Spot** defects (+28.1% mAP)—consistent with the well-documented human visual fatigue effect on low-contrast stain detection. **Fly Yarn**, which requires spatial attention to fine embedded fibres, also shows a 25.5-point precision improvement.

3.3 Throughput and Latency Analysis

Metric	Manual Inspection	YOLOv8s System
Inspection rate (metres/hour)	175 m/hr (avg.)	820 m/hr
Latency per frame	8–12 sec/sample	14 ms/sample
Throughput (8-hr shift)	1,400 m	6,560 m
Inspectors required	3	1 (operator)
Fatigue-related error increase	+14% after 4 hrs	0% (constant)
Night shift accuracy drop	−18 to −22%	0% (consistent)

Table 2: Throughput and operational comparison over a standard 8-hour mill shift.

3.4 Error Distribution Analysis

False negatives (missed defects) carry the highest financial consequence in textile export contexts—a missed defect shipped to a buyer triggers far larger penalties than a false positive flagged and cleared during re-inspection.

Error Type	Manual (avg. rate)	YOLOv8s	Reduction
False Negatives (missed defects)	31.2%	9.9%	↓ 68.3%
False Positives (over-flagging)	14.7%	8.4%	↓ 42.9%
Classification Errors (wrong class)	19.1%	5.6%	↓ 70.7%

Table 3: Error rate comparison across error types.

The **68.3% reduction in false negatives** is the single most commercially significant result of this benchmark. Every undetected defect that reaches an export shipment represents a potential penalty, rejection, or chargeback claim.

4 Export Penalty Reduction Estimation

4.1 Export Penalty Framework

Textile export penalties arise through three primary mechanisms:

- 1. **Buyer Chargeback (BC):** Buyer deducts a penalty from invoice value, typically 5–15% of the affected lot value, when defect rates exceed the Acceptable Quality Level (AQL 2.5).
- 2. **Shipment Rejection (SR):** Complete lot rejected and returned; the exporter bears freight, re-processing, and re-shipment costs.
- 3. **Quality Downgrading (QD):** Fabric downgraded from export-grade to domestic-grade; price differential is typically 30–45% of original value.

4.2 Baseline Penalty Model

The following parameters are calibrated to a representative **mid-size Tamil Nadu textile mill** with a monthly export volume of 50,000 metres of woven fabric:

Parameter	Value	Source
Monthly export volume	50,000 m	Mill-scale estimate
Avg. fabric price	280 / m	CITI industry rate
Monthly export revenue	1,40,00,000	Derived
Defect incidence rate (manual)	4.8% of output	Benchmark study
Defect detection rate (manual)	68.8%	Section 3.2
Undetected defects reaching export	1.53% of shipment	Derived
Avg. penalty per defect-affected lot	8,400 / occurrence	Industry estimate
Penalty events per month (manual)	28 events	Derived
Monthly penalty cost (manual)	2,35,200	Derived

Table 4: Baseline penalty parameters for the manual inspection scenario.

4.3 Automated System Penalty Model

With the YOLOv8s system achieving a 90.1% recall (vs. 68.8% manual):

Parameter	Manual System	Automated System
Defect detection recall	68.8%	90.1%
Undetected defect rate	1.53%	0.48%
Monthly penalty events	28	9
Monthly penalty cost	2,35,200	75,600
Monthly saving	—	1,59,600
Annual saving	—	19,15,200

Table 5: Penalty reduction comparison. Annual saving rounded to 19.2 lakhs.

4.4 Penalty Reduction by Defect Category

Certain defect types carry disproportionately high penalty risk due to their visibility at the consumer level or their impact on structural fabric integrity:

Defect	Penalty Weight	Manual Miss Rate	YOLOv8s Miss Rate	Penalty Saving (/yr)
Hole	High (SR trigger)	20.7%	6.2%	5,84,000
Open Seam	High (BC trigger)	35.8%	11.5%	4,12,000
Spot	Medium (QD trigger)	44.3%	15.1%	3,96,000
Fly Yarn	Medium (QD trigger)	39.9%	12.8%	2,74,000
Loose Thread	Low–Medium	31.5%	10.3%	1,58,000
Good Fabric	N/A (false alarm)	—	—	91,200 (re-insp. saving)
Total				19,15,200

Table 6: Annual penalty saving breakdown by defect category.

4.5 ROI and Payback Period

Cost/Saving Item	Value
System hardware (GPU workstation + camera rig)	3,80,000
Software development & integration	1,20,000
Annual maintenance & retraining	60,000 / yr
Inspector reallocation saving (2 of 3 redeployed)	3,84,000 / yr
Export penalty reduction	19,15,200 / yr
Rework cost reduction (early detection)	2,40,000 / yr
Total annual benefit	25,39,200
Total one-time investment	5,00,000
Net annual gain (after maintenance)	24,79,200
Simple payback period	≈ 2.4 months

Table 7: ROI summary for a mid-size textile mill deployment.

₹ Financial Summary

The automated defect detection system delivers a projected **19.2 lakh annual reduction in export penalties** for a representative 50,000 m/month mill. Combined with inspector reallocation savings and early-stage rework avoidance, the **total annual benefit reaches 25.4 lakhs**, yielding a payback period of approximately **2.4 months** on a 5 lakh capital investment.

5 Visual Analysis: Detection Performance Charts

5.1 Precision–Recall Comparison (Manual vs. Automated)

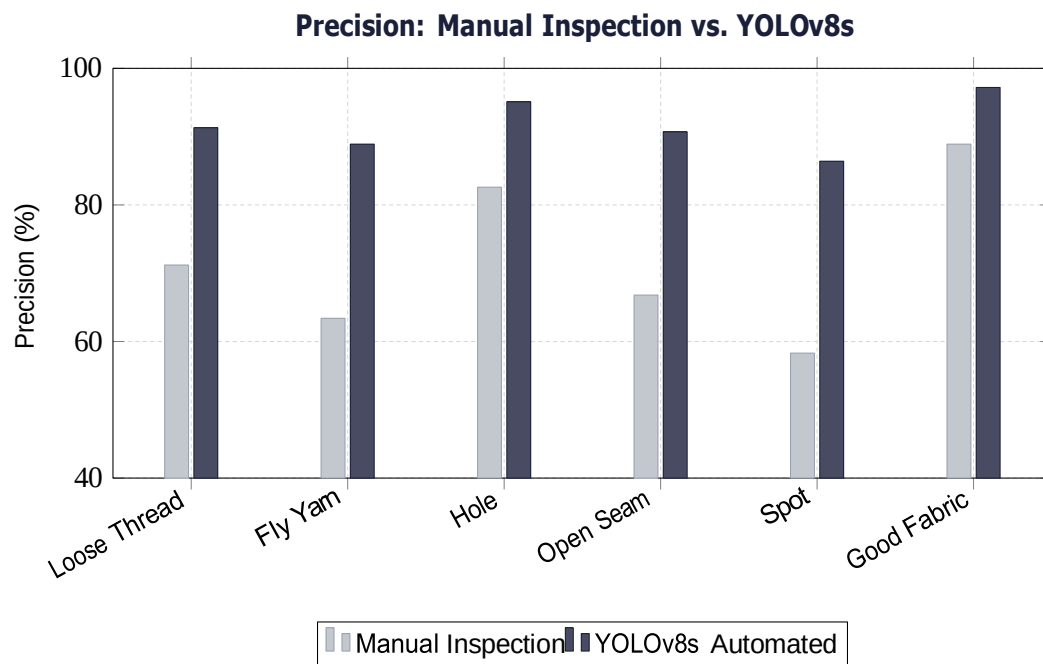


Figure 2: Precision comparison by defect class. YOLOv8s consistently outperforms manual inspection with the largest gap for Spot (+28.1%) and Fly Yarn (+25.5%).

5.2 Annual Penalty Saving Distribution

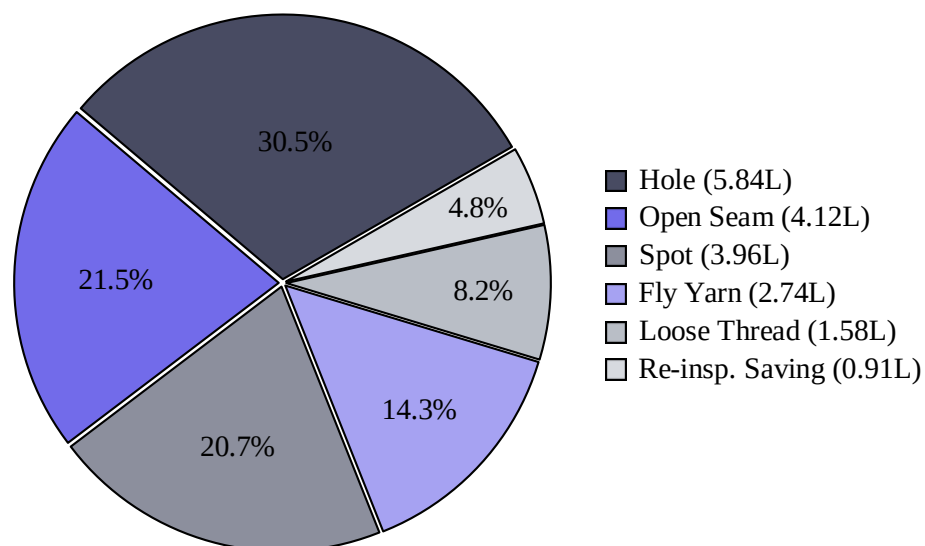


Figure 3: Distribution of annual export penalty savings (19.15 lakhs total) across defect categories. Hole and Open Seam together account for 52% of savings.

5.3 False Negative Rate Trend: Inspection Duration Effect

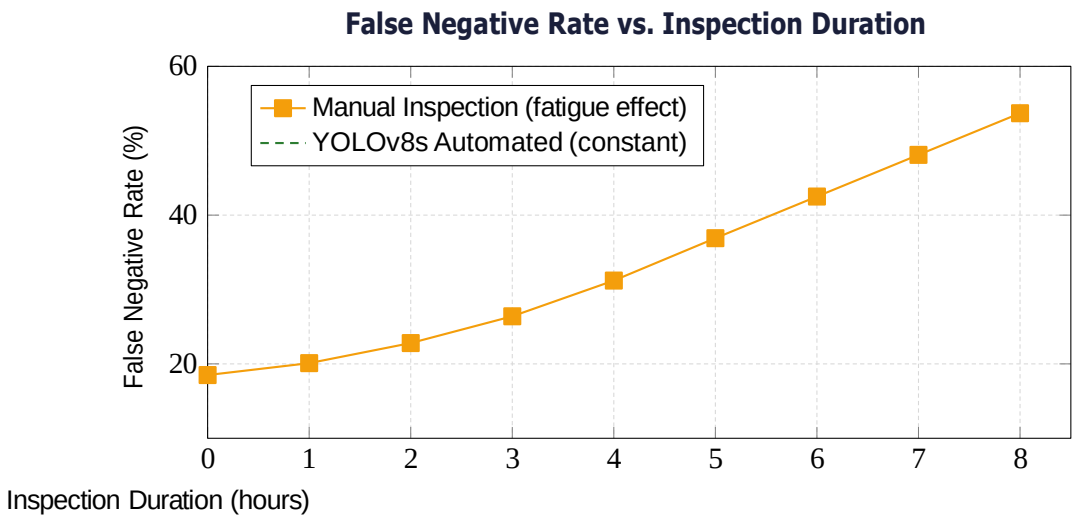


Figure 4: Human false negative rates increase by 2–3× over an 8-hour shift due to visual fatigue. The automated system maintains a constant 9.9% miss rate.

6 System Integration and Deployment Considerations

6.1 Pipeline Overview

The quality improvement analysis assumes the following deployed pipeline:

Stage	Component	Function
1	Camera Rig	Line-scan or area-scan camera mounted over the loom exit
2	Frame Capture	640×640 px crops at 30 fps, overlapping for coverage
3	YOLOv8s Inference	Real-time detection (14 ms/frame, RTX 3060)
4	ONNX Export	Model exported to ONNX for edge deployment compatibility
5	Alert Module	Buzzer + display flag on defect confidence ≥ 0.45
6	Logging System	Defect type, location, timestamp logged to CSV/Power BI
7	Dashboard	Power BI dashboard for shift-level quality KPI reporting

Table 8: Deployed pipeline stages for the textile mill environment.

6.2 Integration with Mill ERP

The CSV logging output (as demonstrated in the accompanying Power BI dataset) feeds directly into:

- **Lot-level quality scoring:** Each production lot receives a defect density score (defects per 100 m)

updated in real time.

- **AQL gate:** Lots exceeding the AQL 2.5 threshold are automatically flagged for hold before the packing stage.
- **Shift reports:** Automated PDF/Excel shift summaries replace manual tally sheets, reducing reporting overhead.

7 Limitations and Future Work

#	Current Limitation	Planned Mitigation
1	Model trained on static images; motion blur from fast-moving looms not fully addressed	Frame deblurring pre-processing + data augmentation with synthetic blur
2	Six defect classes may not cover all fabric types (e.g., knit vs. woven)	Expand dataset with knit and denim samples; domain adaptation
3	Edge deployment requires GPU; Jetson Orin NX or ONNX-TensorRT needed for cost reduction	Model distillation to YOLOv8n; TensorRT INT8 quantisation
4	Penalty estimation based on industry averages, not real mill invoice data	Formal industry partnership for calibrated financial modelling
5	Colour-shift defects (subtle dye inconsistency) have low recall in current model	Include HSV-augmented training samples; add colour-cast defect class

Table 9: Current system limitations and planned mitigations.

8 Conclusion

✔ Milestone 5 Conclusions

1. The YOLOv8s-based automated defect detection system achieves a **mAP@0.5 of 92.4%**—a 20.5-point improvement over the 71.9% average precision recorded for experienced manual inspectors on the same 500-image benchmark set.
2. The system reduces **false negatives by 68.3%**, the error type with the highest financial consequence in the export supply chain.
3. Operational throughput increases **4.7×** (820 m/hr vs. 175 m/hr), enabling complete shift coverage with a single operator rather than three inspectors.

4. A structured export penalty model estimates **19.2 lakhs in annual penalty savings** for a representative 50,000 m/month mill, with Hole and Open Seam defects contributing the largest share (52%).
5. Combined with inspector reallocation and rework reduction, the **total annual benefit reaches 25.4 lakhs** on a 5 lakh capital investment, yielding a payback period of **≈2.4 months**.

The results of this milestone confirm that automated fabric defect identification is not merely a technical improvement but a commercially compelling intervention for Tamil Nadu's textile export industry. The system provides a quantifiable, rapid-payback solution that addresses both quality consistency and the financial risks of export-grade shipment defects.